

Question 1: Word Segmentation and POS Tagging

English (Brown) vs. Spanish (UD Spanish-GSD)

Setup

	English (Brown)	Spanish (UD-GSD)
Train sentences / tokens	45,459 / 804,554	14,186 / 330,548
Word types	41,974	39,748
Type/token ratio	0.052	0.120
Mean token length	4.72 chars	4.80 chars
Dev perplexity (trigram word LM)	1,088.6	2,233.0
OOV rate (test sample)	2.44%	6.66%
Tagset / morphology-aware tagset	11 / 11	16 / 78

Spanish reaches almost the same vocabulary from a quarter of the tokens. Mean token length is identical, so “Spanish words are longer” explains nothing below.

Q1. Where did the two languages differ most?

In segmentation, and in the unknown-word tail - not in tagging.

Measure	English	Spanish	Gap
Segmentation, exact sentences	66.50%	38.75%	+27.75 pp
Segmentation, greedy baseline	66.71%	50.20%	+16.51 pp
Tagging, <i>unknown</i> words only	86.31%	76.53%	+9.78 pp
End to end (segment + tag)	92.56%	86.75%	+5.81 pp
Segmentation, trigram DP (token F1)	95.55%	91.59%	+3.96 pp
Tagging, gold segmentation	96.19%	93.75%	+2.44 pp
Tagging, <i>known</i> words only	96.43%	94.98%	+1.45 pp

Once the words are separated correctly the gap nearly closes (+1.45 pp on known words), so the end-to-end gap is mostly inherited from the segmenter. The cause is sparse statistics, not grammar: Spanish inflection splits each lemma across many forms, doubling perplexity and nearly tripling the OOV rate. Typical failure: `elcielodespejados azul` → `el cielo de espejado es azul`, where a very frequent preposition is available as the prefix of a rarer content word.

Caveat: Spanish is scored on 16 tags against English's 11, which slightly flatters English.

Q2. Did agreement-aware tagging help, or add noise?

It helped in Spanish. In English the question cannot be asked:

Brown carries no FEATS, so the refined tagset is identical to the plain one (11 tags either way). That is a fact about the corpus, not about English.

Accuracy on the refined 78-label set (91.72%) is *not* comparable with the plain tagger's 93.75% - it is a harder decision. Projected back to coarse tags, both models sit on the identical 16-way decision:

Model	Coarse-tag accuracy	Change
Plain trigram HMM	93.75%	-
Morphology-aware HMM, projected	94.23%	+0.48 pp
Plain most-frequent-tag	88.72%	-
Morphology-aware baseline, projected	89.74%	+1.02 pp

So refining the tagset **added no noise**. But the context-free baseline gained *more* than the HMM, so this table alone does not prove the gain comes from agreement modelling. The direct evidence is stronger: in the context (matching determiner, noun) the transition model prefers an agreeing adjective over a clashing one by **12-18×**, and of 2,625 adjacent gold-agreeing pairs it reproduced agreement in **96.46%** (DET+ADJ 99.14%, ADJ+NOUN 98.72%, DET+NOUN 97.68%, NOUN+ADJ 97.15%; weakest NOUN+NOUN 70.83%). End to end this falls to 86.06%, because 305 pairs contain a word the segmenter never recovered.

Both sample strings are fully correct, including the number flip propagating across all four agreeing words:

la/DET-Fem-Sg casa/NOUN-Fem-Sg roja/ADJ-Fem-Sg es/AUX-Sg grande/ADJ-Sg
las/DET-Fem-Pl casas/NOUN-Fem-Pl rojas/ADJ-Fem-Pl son/AUX-Pl grandes/ADJ-Pl

Q3. Segmentation-induced vs. genuine tagging errors

An error is *segmentation-induced* if the gold token's character span was never recovered (no tag decision was made) and *genuine* if the span was right and the tag wrong.

System	Errors	Seg-induced	Genuine	% from seg.	Tag acc. span
English					
Greedy + most-frequent	2,152	1,861	291	86.5%	94.19%
DP segment + most-frequent	661	273	388	41.3%	94.12%
DP segment + HMM tag	511	273	238	53.4%	96.39%
Spanish					
Greedy + most-frequent	4,583	4,162	421	90.8%	92.53%
DP segment + most-frequent	1,576	831	745	52.7%	91.69%
DP segment + HMM tag	1,298	831	467	64.0%	94.79%

With a weak segmenter, segmentation is essentially the *only* error source (86.5% / 90.8%); the baseline tags a correctly segmented token at 94.19% / 92.53%, so nearly all its failure is inherited. Improving the tagger *raises* the segmentation share (41.3% → 53.4%) because it removes genuine errors and cannot touch inherited ones - that number going up is the tagger working. Spanish stays segmentation-dominated (64.0% vs. 53.4%).

The two classes look nothing alike: segmentation-induced - *unfailing* → *un* + *failing*, *dallasbased* → *dallas* + *based*; genuine - *eran* VERB→AUX, *que* PRON→SCONJ. Top confusions: English VERB/NOUN (35+24) and PRT/ADP (17+16); Spanish PROP/NOUN (75+50, unsurprising once case is normalised away) and PRON/DET (42).

Q4. How much better than the simple baselines?

Task	Baseline	Model	Gain	Error reduction
English				
Segmentation (token F1)	66.71%	95.55%	+28.85 pp	86.6%
Segmentation (exact sentences)	16.25%	66.50%	+50.25 pp	60.0%
Tagging (gold segmentation)	93.64%	96.19%	+2.55 pp	40.0%
End to end	68.68%	92.56%	+23.88 pp	76.3%
Spanish				
Segmentation (token F1)	50.20%	91.59%	+41.39 pp	83.1%
Segmentation (exact sentences)	5.75%	38.75%	+33.00 pp	35.0%
Tagging (gold segmentation)	88.72%	93.75%	+5.03 pp	44.6%
End to end	53.21%	86.75%	+33.54 pp	71.7%

Segmentation is where the model pays for itself, removing 83-87% of the baseline's errors: greedy longest-match cannot reconsider a bite, so one wrong match corrupts every boundary after it (only 16.25% / 5.75% of sentences perfect, vs. 66.50% / 38.75%). **Tagging is where the baseline is already strong** - most word types are unambiguous - so +2.55 / +5.03 pp looks small but removes 40-45% of the remaining errors. Both baselines were given every advantage: greedy gets the *full* training vocabulary including hapax words the LM itself discards.

Cost: greedy runs in 0.10 ms/sentence against 92 ms for the DP decoder (English), i.e. three orders of magnitude for 29-41 points of F1. Beam search makes that affordable - at width 8 it matches exact DP to four significant figures (96.53% vs. 96.53%; 92.91% vs. 92.91%) while running 10-20× faster.

Known failures

- brown → NOUN, not ADJ: in the Brown corpus the form is predominantly a proper noun.
- despejado → de + spejado (Q1).
- pueden → AUX where the brief writes VERB: a UD tagset convention, not an error.
- English morphology was never tested - Brown has no FEATS.
- All test figures are 400 sentences per language; differences below roughly half a point should not be read as real.